HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Keet Desktop**, a cross-platform P2P communications application using **React** and the **Pear runtime**, contributing to core feature implementation and performance tuning.
- Implemented state management solutions for the **Keet Desktop** application using **Zustand**, ensuring predictable data flow across complex UI components and improving maintainability.
- Authored comprehensive unit and integration tests for critical application modules within **Keet Desktop**, achieving over **80% code coverage** for new feature development and reducing regression bugs.
- Designed and implemented asynchronous data fetching patterns for **Keet Desktop**, optimizing API interactions and improving the responsiveness of real-time data displays by **~20%**.
- Integrated **WebRTC** APIs within the **Keet Desktop** application to manage real-time audio and video streams, enabling peer-to-peer communication features.
- Containerized the **Keet Desktop** application using **Docker** for consistent development and deployment environments, and managed build pipelines with **Jenkins**.
- Collaborated with design and product teams to translate UI/UX mockups into functional components for **Keet Desktop**, ensuring pixel-perfect implementation and adherence to design specifications.

SKILLS

Frontend Frameworks – React, Preact, TypeScript, JavaScript

State Management – Redux, Redux Toolkit, Zustand

Desktop Development – Electron, Tauri, Pear runtime

P2P & Distributed Systems – BitTorrent-like technology, Decentralized Web, Node.js

Web Technologies – WebRTC, REST APIs, HTML5, CSS3

Development Tools – Docker, Jenkins, Git, Jira, Confluence, Node.js

Testing & Debugging – Unit Testing, Integration Testing, Debugging Tools

PROJECTS

Keet Desktop P2P Communications App (React)

- Developed core features for a cross-platform P2P messaging and file-sharing application using **React** and **TypeScript**, focusing on a peer-to-peer architecture.
- Implemented robust state management with **Zustand** and integrated **WebRTC** for real-time audio/video communication, enhancing user interaction capabilities.
- Wrote extensive unit and integration tests to ensure application stability and maintainability, contributing to a smooth user experience for early adopters.

Pear Runtime Integration & Optimization

- Engineered direct application deployment mechanisms from developer machines to users via the **Pear runtime**, mimicking **BitTorrent** principles for data replication and scalability.
- Optimized rendering performance and memory management for the **Keet Desktop** application running on the **Pear runtime**, reducing CPU load by **~15%** during peak usage.
- Containerized the P2P application components using **Docker** to facilitate seamless deployment across macOS, Windows, and Linux environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)